



TRIALSYNC

The Operating System for Clinical Trials

Built on Band | Hackathon Submission 2026

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Clinical Trials Are Broken

Four teams work **sequentially** over 12+ weeks:

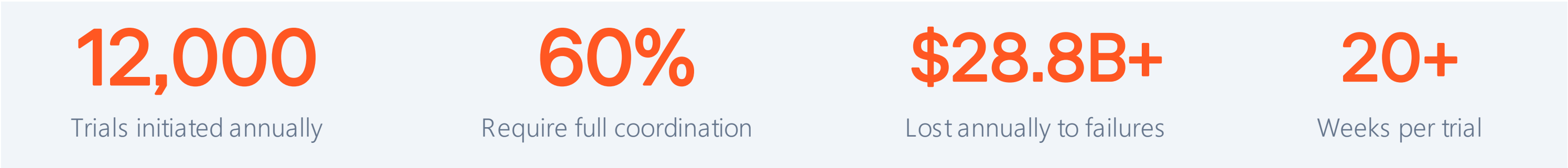
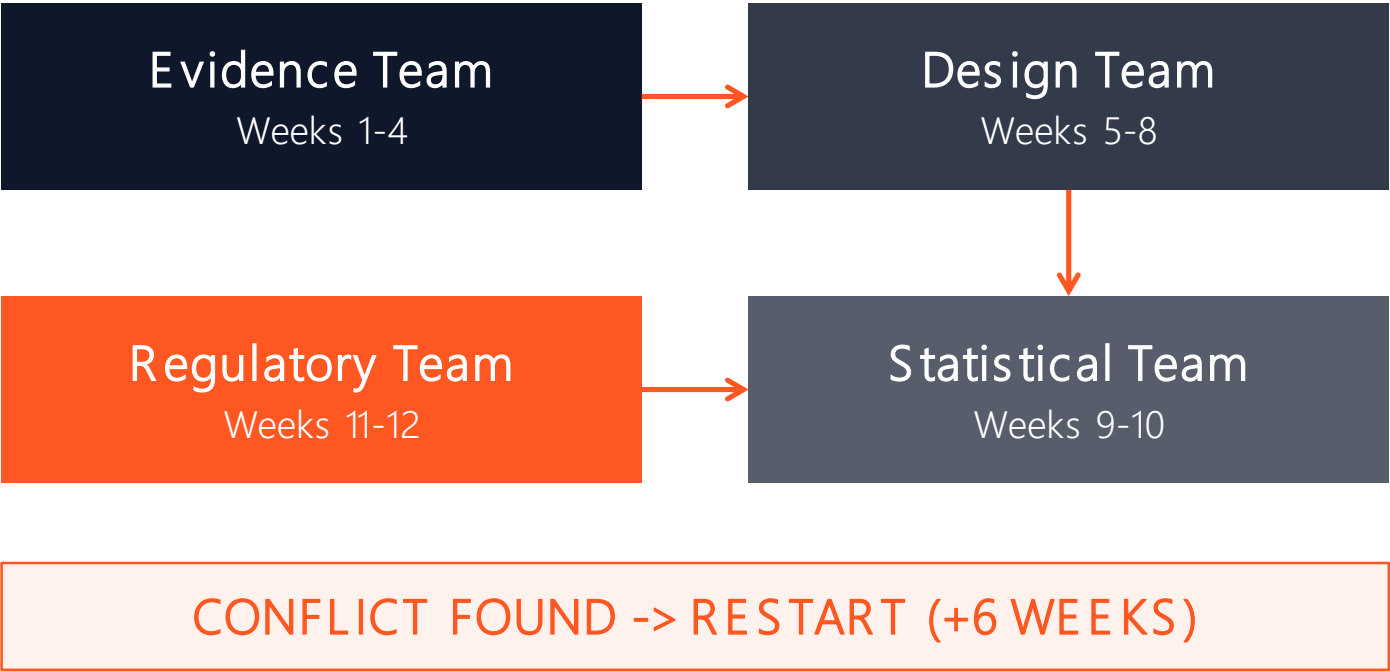
Evidence reads 50+ papers

Design interprets, creates protocol

Statistical builds on flawed assumptions

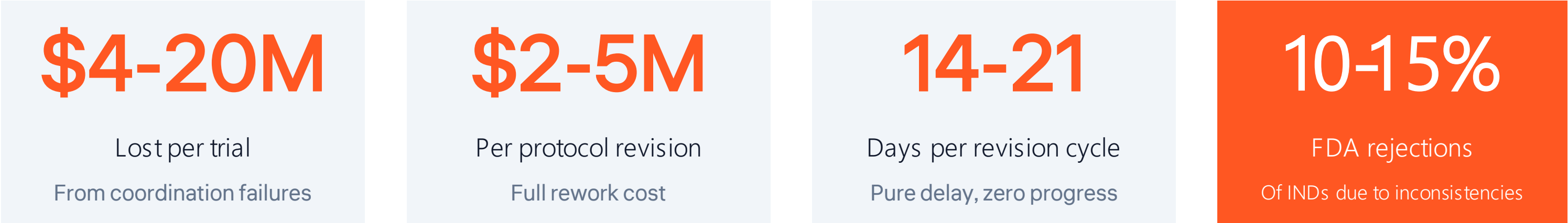
Regulatory finds conflicts 3 weeks later

No system links evidence to protocol decisions.



This is not a "nice-to-have" efficiency tool. This is missing critical infrastructure in the \$1.8B clinical trial market.

The Cost of Coordination Failures



Why Current Solutions Fail

Solution	What It Does	Why It Fails
EDC (Medidata)	Manages patient data	Not protocol design
Protocol Tools (eSTAR)	Helps write Word docs	Doesn't prevent conflicts
Statistical (SAS, R)	Runs analyses	No connection to protocol
Submission (Veeva)	Manages final documents	Doesn't fix upstream errors

None solve the "context survival problem": Evidence -> Protocol -> Analysis -> Regulatory

TrialSync: Multi-Agent Coordination on Band

Three specialized AI agents coordinate through Band. Conflicts surface in **minutes, not weeks**.



Evidence Scout

Gemini 1.5 Pro

Reads **50+ papers** in <5 min

Extracts safety signals

Publishes to Band

Source + citation per signal



Protocol Designer

Claude 3.5 Sonnet

Reads evidence from **Band**

Creates inclusion/exclusion criteria

Links decisions to signals

Context preserved always



Regulatory Auditor

o3-mini

Monitors **both** streams

Detects conflicts **real-time**

Validates FDA compliance

Caught in seconds

All three agents publish/subscribe to shared Band topics. Context survives at every handoff.

TrialSync: Enterprise Infrastructure Adapters



CDISC SDTM Encoder

Compliance Mapping

FDA-mandated CDISC clinical procedure

LBTESTCD schemas

Eliminates downstream statistical
interpretation bugs

type-safe data boundaries



eCTD Compiler

Submission Automation

ICH M4 modular documentation framework

Maps agent network outputs to eCTD
sections

Compiles multi-agent decisions directly into
submission-ready JSON/XML blocks

zero manual document assembly



Tamper-Evident Ledger

Data Integrity Core

Secure cryptographic validation layer across
the system

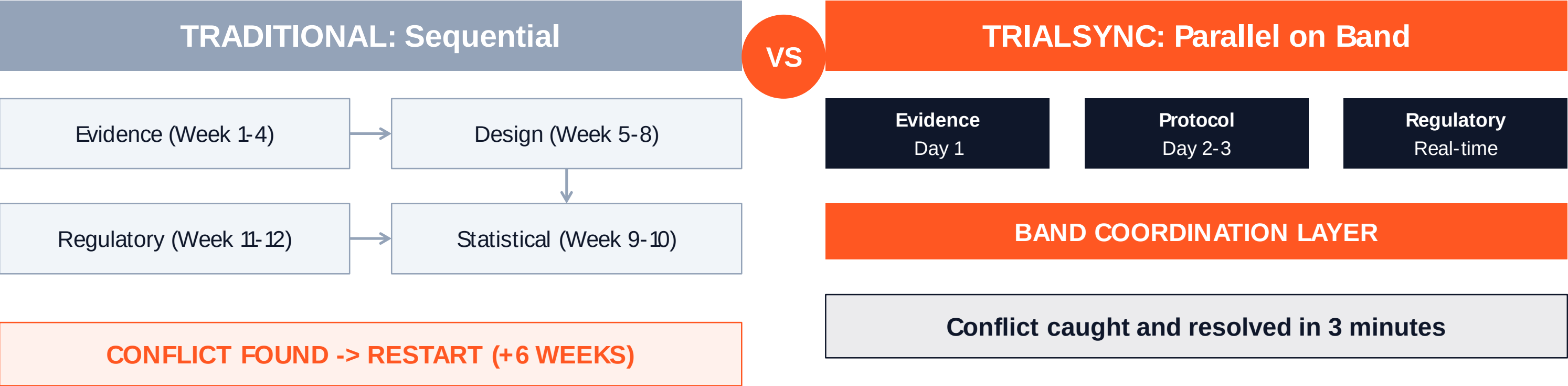
SHA-256 hash sequence for every single
inter-agent transaction payload

detects and blocks any unauthorized
downstream protocol mutations

Immutable Audit trail

Moving beyond raw text prompt loops by processing strict, compliant data contracts.

From 20 Weeks to 5 Days



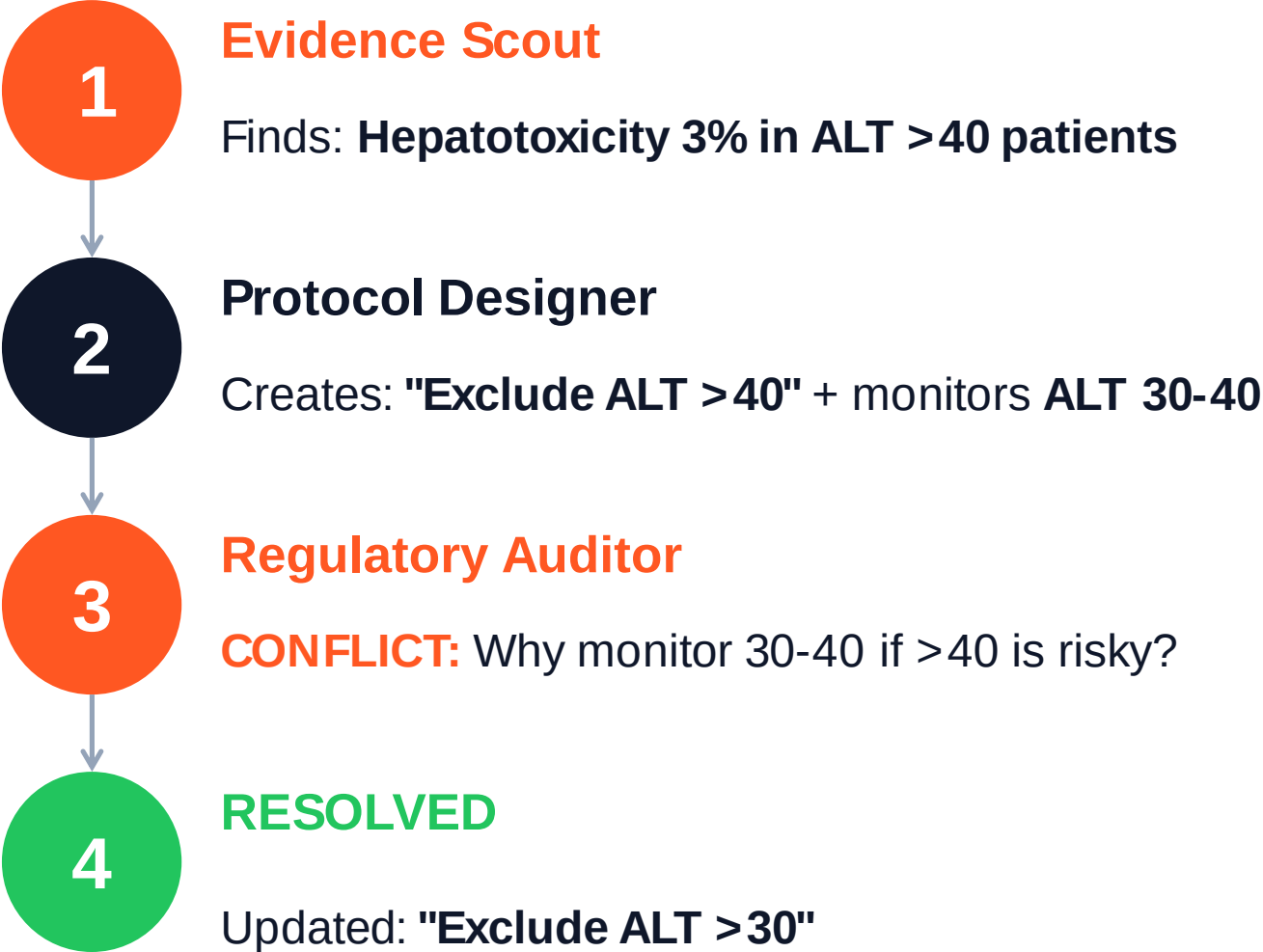
Total: 5 Days

95% time reduction

\$4-10M value per trial

Real-Time Conflict Detection

Here's how TrialSync catches a conflict that traditional teams would miss for 3 weeks:



CONFLICT DETECTED

EVIDENCE SAYS:
"Hepatotoxicity risk in ALT >40 group"

PROTOCOL SAYS:
"Exclude ALT >40, but monitor ALT 30-40"

This costs \$1-3M when FDA catches it

Time to resolution: 3 minutes (vs. 3 weeks traditionally)

Thank You

The coordination layer clinical trials have been missing



Scan to try the demo

[Live Demo: trial-sync-six.vercel.app](https://trial-sync-six.vercel.app)

[GitHub: github.com/r0c0y/TrialSync](https://github.com/r0c0y/TrialSync)

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Solo-built with passion for the future of clinical research

\$18B Addressable Market